TV5: LOCKING, BYZANTINE HANDLING AND TEST EVIDENCE

Implementation notes | Tendermint-style BFT | n = 3f + 1 | Reproducible Python tests

1. Scope and safety objective

TV5 guards the boundary between untrusted consensus messages and validator state. **VoteTracker** validates, de-duplicates and tallies votes; **LockManager** enforces local signing and PoLC-based lock changes; the Byzantine simulator creates bounded faults; and the deterministic logger emits byte-stable evidence. The safety invariant is that honest nodes never finalize two blocks at the same height.

2. Seven voting and locking rules

Rule	Enforced condition	Implementation
1	One PREVOTE per validator and (H, R).	Local slot keyed by height, round and phase.
2	One PRECOMMIT per validator and (H, R).	PRECOMMIT uses an independent phase slot.
3	Non-NIL PRECOMMIT requires matching PREVOTE quorum.	can_precommit() checks a stored PoLC.
4	2f + 1 PREVOTE(B), B != NIL, locks B at round R.	observe_quorum() validates and records lock.
5	Changing block requires locked_round < polka_round < proposal_round.	can_prevote() rejects stale/same-round PoLC.
6	Reject invalid voter, coordinates, hash, chain or signature domain.	Optional verification uses VOTE:<chain_id>.
7	Duplicates count once; conflicts are equivocation.	First vote stays; conflicting pair is evidence.

3. Quorum, locking and PoLC

Votes are indexed by (height, round, phase) and tallied per block hash. A quorum contains distinct validators for one block. PoLC votes must agree on height, round, PREVOTE phase and block hash. A new height clears the lock while retaining the local signing history for audit.

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Safety-critical ordering: locked_round < polka_round < proposal_round
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4. Implementation notes

Exact retransmissions are idempotent; conflicting duplicates never replace the first admitted vote. Signature payloads use stable sorted JSON and the VOTE:<chain_id> domain. Logger timestamps use simulated time, sorted compact JSON, UTF-8, LF and one final newline. These choices prevent packet order, locale and platform formatting from changing the tally or evidence hash.

5. Byzantine handling

Attack	Injected behavior	Expected defense
Equivocation	Vote A for one network view and B for another.	Count one vote; retain conflict evidence.
Invalid state	Propose a deterministic wrong state commitment.	Execution/ledger validation rejects it.
Replay	Repeat an old packet or vote unchanged.	Vote/transaction handling stays idempotent.
Silence	Withhold all outbound messages.	Timeout and round change preserve liveness.

6. Consolidated T1-T8 results

Status is limited to evidence present on this branch. PARTIAL means a supporting unit or component test passes but the named full-stack E2E scenario still requires TV1-TV4 integration.

Test	Required scenario	Evidence on TV5 branch	Status
T1	>=8 nodes: same finalized chain/state	Needs integrated node/state stack.	**NOT RUN**
T2	Duplicate/reordered packets	Vote de-dup PASS; execution/reorder E2E pending.	**PARTIAL**
T3	Invalid signature/domain	Vote-domain rejection PASS; transaction E2E pending.	**PARTIAL**
T4	Transaction replay	Replay primitive PASS; ledger E2E pending.	**PARTIAL**
T5	Drop/delay: safety and liveness	Needs integrated network scenarios.	**NOT RUN**
T6	Proposer failure and recovery	Needs integrated consensus FSM.	**NOT RUN**
T7	Byzantine equivocation	One common block; no conflicting finalize.	**PASS**
T8	Deterministic execution	Identical logs, block set and state hash.	**PASS**

7. Reproducibility evidence

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Log SHA-256 (run 1 = run 2): 3c5c2ac64672dc7a25746c1d5220d10cbe34f18277423def81a02bd2b9560afa
Final state hash: 654698a006e8cdee6c249d932aa088ddbbc1cad4098284e52fb241b61f3b2f34
Repository suite at publication: 34/34 PASS; Ruff: no findings.
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8. References

[1] L. Lamport, R. Shostak, M. Pease, *The Byzantine Generals Problem*, ACM TOPLAS 4(3), 1982. doi:10.1145/357172.357176
[2] M. Castro, B. Liskov, *Practical Byzantine Fault Tolerance*, OSDI, 1999. usenix.org/conference/osdi-99/practical-byzantine-fault-tolerance
[3] E. Buchman, J. Kwon, Z. Milosevic, *The latest gossip on BFT consensus*, arXiv:1807.04938, 2018.